The Exponential Distribution

2026-04-17

Introduction

The exponential distribution models waiting times between events in a Poisson process with constant rate. It is the only continuous distribution with the memoryless property, and it is the simplest parametric model in survival and reliability analysis.

Prerequisites

Continuous random variables and the Poisson process.

Theory

A non-negative random variable X is **exponential with rate** $\lambda > 0$ if

$$f(x) = \lambda e^{-\lambda x}, \qquad x \geq 0.$$

CDF: $F(x) = 1 - e^{-\lambda x}$. Survival function: $S(x) = e^{-\lambda x}$.

Moments:

- $E[X] = 1/\lambda$ (the mean waiting time).
- $\text{Var}(X) = 1/\lambda^2$.

Memoryless property: $P(X > s + t \mid X > s) = P(X > t)$. The probability of waiting another t units, given we have already waited s, is the same as the unconditional probability of waiting t.

Constant hazard: $h(x) = f(x)/S(x) = \lambda$. The instantaneous event rate does not depend on how long the object has already survived.

Relationship to Poisson: in a Poisson process with rate λ, inter-arrival times are iid exponential with rate λ.

Scale vs rate: some R functions use rate (λ), others use scale ($1/\lambda$). Check the documentation.

Assumptions

Constant hazard (no ageing / no improvement). Violations motivate the Weibull or log-normal distributions.

R Implementation

```r
lambda <- 0.5

X <- rexp(1e5, rate = lambda)
c(mean = mean(X), theoretical = 1 / lambda,
  var  = var(X),  theoretical_var = 1 / lambda^2)

# Memoryless property
P_past_2 <- mean(X > 2)
```

```r
P_past_5_given_2 <- mean(X[X > 2] > 5)         # P(X > 5 | X > 2)
P_past_3 <- mean(X > 3)                         # should equal above by memorylessness
c(conditional = P_past_5_given_2, unconditional = P_past_3)

# Constant hazard
breaks <- seq(0, 10, by = 1)
hazards <- sapply(seq_len(length(breaks) - 1), function(i) {
  in_bin <- X >= breaks[i] & X < breaks[i + 1]
  sum(in_bin) / sum(X >= breaks[i])
})
round(hazards, 3)
```

Output & Results

```
    mean theoretical       var  theoretical_var
   1.996         2.0     3.985              4.0

  conditional  unconditional
       0.2232         0.2229

 [1] 0.394 0.392 0.392 0.400 0.391 0.395 0.394 0.389 0.399 0.394
```

Empirical mean and variance match theoretical; memoryless property holds; empirical hazard is constant at $\approx 1 - e^{-\lambda}$ per unit interval.

Interpretation

Exponential survival is the simplest hypothesis one can test in reliability and survival. When observed hazard is non-constant, report the Weibull or a flexible parametric model instead.

Practical Tips

- `rexp(n, rate)` in R uses the rate parameterisation; `scale = 1/rate`.
- Log-transform to compress the heavy right tail for visualisation.
- Constant hazard is a strong and testable assumption; plot $-\log S(t)$ against t; linearity indicates exponentiality.
- For right-censored data, use survival-analysis tools (`survreg(..., dist = "exponential")`) rather than naive `rexp`.
- Exponential is the discrete geometric's continuous analogue; both share memorylessness.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots